DAY 1 / THEORY / 1 HOUR

Introduction to Networking Hardware

A simple, picture-based guide to the devices that bring the internet into your home: **modem, router, switch, access point** and **network cables**.

The big idea in one minute

Your internet company (ISP) sends internet to your home through a cable. A **modem** receives it. A **router** shares it with all your devices. A **switch** adds more cable ports, and an **access point** adds Wi-Fi. **Cables** carry the data between them. Think of it like water supply: the modem is the main pipe entering the house, the router is the tap that splits water to every room, and the cables are the pipes themselves.

How data travels into your home (the router is highlighted because it is the brain of the setup)

Quick glossary (easy words)

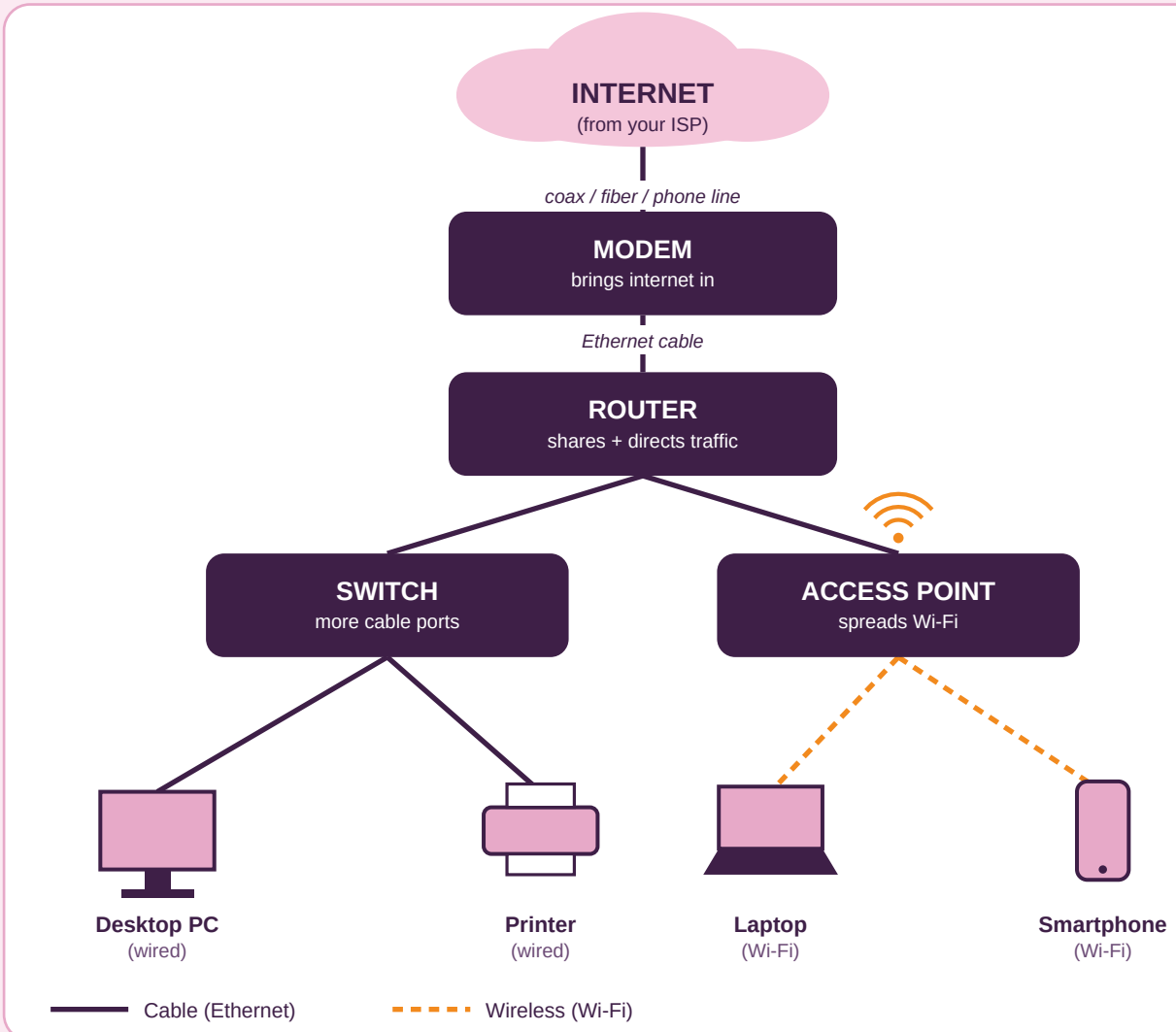
ISP	Internet Service Provider, the company that sells you internet (for example Jio, Airtel, BSNL).
LAN	Local Area Network, your devices connected together inside one home or office.
WAN	Wide Area Network, the big outside network, usually the internet.
IP address	A number that identifies a device on a network, like a house address.
Port	The socket on a device where a cable plugs in.
Wi-Fi	Wireless way of connecting devices using radio waves instead of cables.

What is in this assignment

1. Labeled diagram of a home Wi-Fi setup
2. Function and real-life example of each device
3. Three types of network cables with image sources
4. Router vs switch comparison table

1 Diagram: Typical Home Wi-Fi Setup

Modem, router, switch, access point and four connected devices



How to read the diagram

- **Modem** receives the signal from the ISP and turns it into data your network understands.
- **Router** gives every device its own address and sends data to the right place.
- **Switch** connects wired devices (PC, printer) to the router through cable.
- **Access point** lets wireless devices (laptop, smartphone) join the network using Wi-Fi.

Good to know: In most homes, one box (the "Wi-Fi router" from your ISP) combines the modem, router, a small 4-port switch and an access point. This diagram shows them separately so you can learn what each part does.

2 Main Function and Real-Life Example of Each Device

What it does, in one simple line, and where you will see it

Device	Main function (simple words)	Real-life example
Modem	Connects your home to the ISP. It converts the signal from the ISP line (cable, fiber or phone line) into digital data your devices can use, and back again.	The box your internet company installs where the cable enters your home, for example an ISP fiber/ONT box or a cable modem.
Router	Connects your home network to the internet and decides where each piece of data should go. It also gives IP addresses to devices and usually provides Wi-Fi.	Router: at home to share one internet connection with phones, laptops and TV. Also used in offices and cafes.
Switch	Connects many wired devices in the same local network and sends data only to the device that needs it.	Switch: in a school computer lab or office, joining 24 PCs together; also inside a server room.
Access point (AP)	Creates a Wi-Fi network so wireless devices can join a wired network. It extends Wi-Fi coverage to places the router cannot reach.	Access point: on the ceiling of a college library, mall or large office so Wi-Fi works on every floor.
Network cable	Carries data signals between devices as electrical pulses (copper) or light (fiber). It is the physical path of the network.	Network cable: the Ethernet cable joining your router to a desktop PC or smart TV.

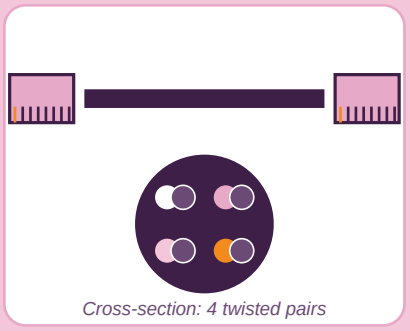
Easy way to remember

Modem Door to the internet	**Router** Traffic police between networks	**Switch** Power strip for cables	**AP** Wi-Fi antenna	**Cable** The road for data

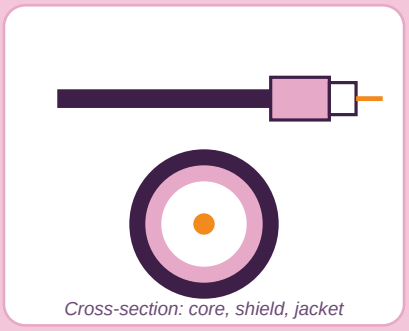
3 Three Types of Network Cables

Ethernet (twisted pair), coaxial and fiber optic

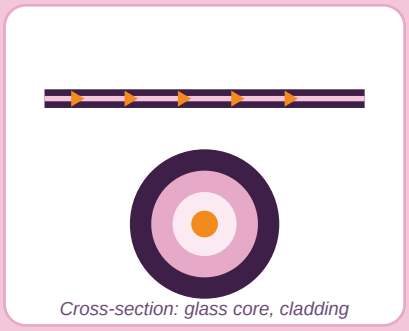
Ethernet (twisted pair)	Coaxial cable	Fiber optic cable



Cross-section: 4 twisted pairs



Cross-section: core, shield, jacket



Cross-section: glass core, cladding

Drawn illustrations (not photographs), showing the cable and the inside of each type. Real photos are available at the source links below.

Cable	Where it is commonly used (one line)	Image source link
Ethernet (Cat5e / Cat6)	Connects computers, TVs and routers to a wired LAN inside homes, schools and offices.	https://en.wikipedia.org/wiki/Category_5_cable
Coaxial	Carries cable-TV and cable-internet signals from the street to your home and to set-top boxes.	https://en.wikipedia.org/wiki/Coaxial_cable
Fiber optic	Sends internet as light over long distances: ISP backbones, FTTH broadband and data centers.	https://en.wikipedia.org/wiki/Optical_fiber_cable

Quick comparison

	Ethernet	Coaxial	Fiber optic
Carries	Electrical signals	Electrical signals	Light pulses
Core material	Copper wires	Single copper core	Glass or plastic
Speed / distance	Good, up to 100 m	Good, medium range	Fastest, very long range

Note for submission: if your teacher needs real photographs, open the links above, or take photos of the cables at home or in your lab, and attach them to this page.

4 Router vs Switch: Comparison Table

Six key differences including main use, location and data handling

Point of difference	Router	Switch
Main use	Connects **different networks** together, such as your home network and the internet.	Connects **devices inside one network**, such as PCs and printers in a lab.
Typical location	At the edge of the network, where the internet enters a home or office.	Inside the network, between the router and the wired devices (office floor, server room, lab).
How it handles data	Reads the **IP address** of each packet and chooses the best path (routing).	Reads the **MAC address** and forwards data only to the correct port (switching).
Works at (OSI layer)	Layer 3: Network layer.	Layer 2: Data Link layer (most basic switches).
IP address handling	Assigns IP addresses to devices (DHCP) and can share one public IP using NAT.	Does not assign IP addresses; it only moves data between ports.
Extra features	Wi-Fi, firewall, security, parental controls (in home routers).	Usually none; more ports (8, 16, 24, 48). Managed types add VLANs.
Real-life example	The Wi-Fi box from your ISP at home.	An 8-port box in a computer lab joining all PCs.

In one sentence

A **switch** connects the devices *inside* your network, while a **router** connects your network to *other networks* like the internet.

Day 1 key takeaways

- The **modem** brings the internet in; the **router** shares it.
- A **switch** adds wired ports; an **access point** adds Wi-Fi.
- **Cables** (Ethernet, coaxial, fiber) are the roads that carry data.
- Home Wi-Fi routers usually combine all four devices in one box.